



Royal
Glass



Duo

Glass Balustrade System

Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Double Disc Glass Balustrade
Supplied to:	Auckland Council
Client / designers name:	
Description of work: <i>E.g. New dwelling</i>	
Address of property:	

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☐ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ <i>please specify</i>	
Location of deck / balcony / stairs:	☐ Internal	☐ External	☐ <i>please specify</i>			
Deck / balcony / stairs:	☐ New	☐ Existing	Cladding:	☐ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☐ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer
Thickness:	(mm)	Width of pane:	0.5-1.8 (m)
Height above floor:	(m)	Height above fixing:	(m)
Note: If barrier extends more than 1.2m above fixing point, test reports must be provided			
Supplier:	Royal Glass		

PART 4: BARRIER DETAILS

System name:	Double Disc Glass Balustrade
Drawings (ref page number)	
Engineers name:	PNP Engineering
Hardware (type):	316 Stainless Steel

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES (tick all boxes that apply)

- | | |
|--------------------------|---|
| ☐ | Continuous interlinking rail (show how fixed to building and each pane of glass) |
| ☐ | Continuous interlinking top cap (show how fixed to building and each pane of glass) |
| ☐ | Clamps, brackets or point fixings (show how fixed to building and each pane of glass) |
| ☐ | Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass) |
| ☐ | Glass with stiff interlayer |
| ☐ | Two or three glass edges supported by continuous clamps |
| ☐ | Structural post to fix interlinking rail to deck |
| ☐ | Other (specify) |

Engineers name:

PNP Engineering

Drawings (ref page number)

Comments:

PART 6: COVER SHEET COMPLETED BY

Name:

Roxy Huang

Signature:



Date:

Contact number:

0800 769 254

Email:

info@royalglass.co.nz



Duo is a strong and durable point-fixed frameless glass balustrade system. It features two anchors that are securely bolted through the glass panels. This provides extra support and rigidity. The compactness of the bolts and their covers reduces the visible hardware, ensuring a modern style without compromising the security and reliability of your balustrades.

1 Choose your glass configuration



Toughened Glass **SAFE**

Toughened glass have greater strength and safety features than float glass, allowing the glass to break with dull edges.

Interlinking rail and gap clamps may be required*



Laminated Glass **SAFER**

Laminated glass have EVA films between 2 or more layers of toughened glass creating a single panel, guaranteeing that if the glass shatters, it remains bonded together.



SentryGlas™ **SAFEST**

SentryGlas™ have interlayers in the glass that is significantly stronger and stiffer than traditional laminates, ensuring minimal deflection in case of glass breakage.

2 Choose your glass type



Standard

Has a slight green tint from standard glass attributes



Low-Iron

Provides a clearer and more transparent view



Frosted

Gives superior privacy and security



Tinted

Minimises sun exposure and provides privacy

Your balustrades may consist of the following hardware fixings:



A Interlinking Rail



B Glass Clamp



C Rail Joiner



D Wall Clamp



E Anchor



Disc

Square

Would you like guidance on the most suitable option for your property? Our team is ready to assist you.

Give us a call at 0800 769 254 or send us an email at info@royalglass.co.nz

3 Choose your fixing colour



Chrome Finish



Satin Finish



Powdercoated

P & P CONSULTING ENGINEERS LTD

Civil and Structural Engineering

Mr P. Prakash (Director)

B.E. (Civil), CPEng, MIPENZ

Mr J. Dela Cruz

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Ref: 20/059/dd.ds

21/06/2024

ROYAL GLASS LTD

CANTILEVER GLASS BALUSTRADES SUPPORTED BY DOUBLE DISCS

DESIGN CALCULATIONS SUMMARY

INDEX	PAGE
1. <u>Producer Statement - Design</u>	1
2. <u>Occupancy Types</u>	2
The balustrade has been tested to comply with domestic barriers serving one or more dwelling including balcony edges (occupancy types A & C3 of AS/NZS 1170).	
3. <u>Test Results</u>	3-5
Refer test results on pages 3 – 5. 12 mm toughened glass is assessed to be sufficient for max. height 1.13 m from centre of top disc to top of glass. Glass to have continuous top rail 50 x 1.5 CHS or 25x21 RHS	
4. <u>Wind Load Assessment</u>	6
The balustrade is sufficient for up to “very high” wind zone.	
5. <u>Fixings</u>	6
Refer to summary drawing ENG 01.	
6. <u>Interlinking Rail</u>	7-10
Use 50.8 OD x 1.5 mm thick CHS rail or 25 x 21 SHS rail.	
7. <u>Summary</u>	11
Refer summary sketch drawing ENG 01 & ENG 04.	

Notes:

1. Any parts of the structure which are not covered by the specific design included with these calculations must comply either with the New Zealand Building Code or specific design as detailed by others. Any exceptions to this should be referred back to this Design Office.
2. The above calculations include structural work for which a Building Consent must be obtained prior to building. It is the Owner's responsibility to obtain all necessary consents.
3. It is assumed that the strength and stiffness of the substrate is sufficient to adequately resist the balustrade loads – this must be confirmed for each installation situation.
4. This design assumes that all the specified members are suitably protected from excess moisture in accordance with Section E1, E2 and E3 of the Building Code. All timber, steelwork, bolts and fasteners to be corrosion protected in accordance with the requirements of NZS 3604:2011 Chapter 4, Durability.

SUPPLIER NAME: ROYAL GLASS
NAME:
ADDRESS:
DATE:
CONSENT AUTHORITY:

engineering
new zealand



New Zealand
Institute of Architects
Incorporated



THIS PS1 DOES NOT CONSTITUTE A SITE SPECIFIC REVIEW BY THE ENGINEERING COMPANY BELOW AND IS ONLY APPLICABLE FOR STANDARD FIXINGS AND DESIGN PARAMETERS AS DETAILED IN THE RELEVANT MANUAL

Building Code Clause(s) B1, F2, F4

PRODUCER STATEMENT – PS1 – DESIGN

(Guidance on use of Producer Statements (formerly page 2) is available at www.engineeringnz.org)

ISSUED BY: P & P CONSULTING ENGINEERS LTD

(Design Firm)

TO: ROYAL GLASS LTD

(Owner/Developer)

TO BE SUPPLIED TO: VARIOUS COUNCIL

(Building Consent Authority)

IN RESPECT OF: GLASS BALUSTRADE & POOL FENCE SUPPORTED BY DOUBLE DISCS, CAPPING RAILS

(Description of Building Work)

AT: VARIOUS

(Address)

Town/City: LOT DP SO
(Address)

We have been engaged by the owner/developer referred to above to provide:

STRUCTURAL DESIGN

(Extent of Engagement)

services in respect of the requirements of Clause(s) B1, F2, F4 of the Building Code for:

☐ All or ☒ Part only (as specified in the attachment to this statement), of the proposed building work.

The design carried out by us has been prepared in accordance with:

☒ Compliance Documents issued by the Ministry of Business, Innovation & Employment B1/VM1, F2/AS1, F4/AS1 or
(verification method/acceptable solution)

☐ Alternative solution as per the attached schedule

The proposed building work covered by this producer statement is described on the drawings titled:

FRAMELESS GLASS BALUSTRADE & POOL FENCE and numbered 20/059 ENG 01 - ENG 07
together with the specification, and other documents set out in the schedule attached to this statement.

On behalf of the Design Firm, and subject to:

- (i) Site verification of the following design assumptions REFER NOTES AT END OF DESIGN SUMMARY
(ii) All proprietary products meeting their performance specification requirements;

I believe on reasonable grounds that a) the building, if constructed in accordance with the drawings, specifications, and other documents provided or listed in the attached schedule, will comply with the relevant provisions of the Building Code and that b), the persons who have undertaken the design have the necessary competency to do so. I also recommend the following level of construction monitoring/observation:

☐ CM1 ☐ CM2 ☐ CM3 ☐ CM4 ☐ CM5 (Engineering Categories) or ☒ as per agreement with owner/developer (Architectural)

I, PARMIL PRAKASH am: ☒ CPEng 251801 # ☐ Reg Arch #
(Name of Design Professional)

I am a member of: ☒ Engineering New Zealand ☐ NZIA and hold the following qualifications: BE (Civil), CPEng

The Design Firm issuing this statement holds a current policy of Professional Indemnity Insurance no less than \$200,000*.

The Design Firm is a member of ACENZ: ☐

SIGNED BY PARMIL PRAKASH (Signature)
(Name of Design Professional)

ON BEHALF OF P & P CONSULTING ENGINEERS LTD Date: 31/10/2024
(Design Firm)

Note: This statement shall only be relied upon by the Building Consent Authority named above. Liability under this statement accrues to the Design Firm only. The total maximum amount of damages payable arising from this statement and all other statements provided to the Building Consent Authority in relation to this building work, whether in contract, tort or otherwise (including negligence), is limited to the sum of \$200,000*.

This form is to accompany Form 2 of the Building (Forms) Regulations 2004 for the application of a Building Consent.
THIS FORM AND ITS CONDITIONS ARE COPYRIGHT TO ACENZ, ENGINEERING NEW ZEALAND AND NZIA

P & P CONSULTING ENGINEERS LTD

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Ref: 20/059

TO WHOM IT MAY CONCERN

Dear Sir/Madam,

B2 COMPLIANCE

Producer Statement for Design-PS1 for Clause B2 of the Building Code – Structural Durability.

We are not able to provide this because there is no effective verification method for B2 contained within the Building Code.

However we can confirm that for the structural elements shown on our documentation:

Timber

Timber treatment has been selected in accordance with Table 1A of B2/AS1.

Concrete

Concrete cover have been selected in accordance with NZS3101:2006, Part 1 Section 3.

Mild Steel

Steel protection has been specified in accordance with the “Guide to the protection of structural steel against atmospheric corrosion by the use of protective coatings” AS/NZS3212. We Note that this is on a time to first maintenance basis.

We trust this provides the information that you are seeking.

Yours faithfully,



Mr P Prakash
Director
P & P Consulting Engineers Limited

ASSESSMENT OF TESTING OF CANTILEVER GLASS BALUSTRADES

USING DOUBLE DISC SUPPORT SYSTEM

2. OCCUPANCY TYPES - DOMESTIC & RESIDENTIAL ONLY

The glass and support channel was tested to comply with the following domestic load cases taken from AS/NZS 1170.1:2002 and the DBH Practice Advisory 10:

Occupancy Type	Refer Table 3.3 of AS/NZS 1170:	Maximum Design Loads		
A, C3 (Residential Only)	Domestic Barriers for One or More Dwellings including Balcony Edges (NOT subject to Over Crowding)	0.75 kN/m (75 kg/m)	0.6 kN (60 kg) Anywhere	1 kPa (100 kg/m ²) Infill

3. TEST RESULTS

As noted above, the test loads were determined from Table 3 of AS/NZS 1170:2002

3 samples were tested which passed. Refer to test results attached on pages 3 -5

Based on above testing, 12 mm toughened safety glass is sufficient for the occupancy types noted above.

Deflection:

Take allowable deflection as height / 30.

In this case, the height = 1.13m (above upper row of fixing)

Hence, allowable deflection = 36 mm

From the test results, the deflection at serviceability loads are summarised below:

SLS load	Deflection
0.6 kN at corner	25 mm
1.6 kN UDL	34 mm

As can be seen, the deflections are within the normal acceptable limits

3

Testing 1

Point :	
Stress(kg)	Distance(mm)
0	0
20	8
40	17
60	25
80	31
100	39
110	52
Face:	
Stress(kg)	Distance(mm)
0	0
20	2
40	6
60	10
80	17
100	22
120	25
140	30
160	36
180	43
200	48
220	53
240	59
260	64
280	69
300	75
320	80
340	85
360	92

Line:	
Stress(kg)	Distance(mm)
0	0
20	5
40	9
60	13
80	17
100	22
120	26
140	29
160	34
180	43
200	43
220	44
240	50
260	54
280	59
300	63
320	fracture

Testing 2

Point :	
Stress(kg)	Distance(mm)
0	0
20	8
40	17
60	25
80	31
100	39
110	53
Face:	
Stress(kg)	Distance(mm)
0	0
20	2
40	6
60	10
80	17
100	22
120	25
140	30
160	36
180	43
200	48
220	53
240	59
260	64
280	69
300	75
320	80
340	85
360	92

Line:	
Stress(kg)	Distance(mm)
0	0
20	5
40	9
60	13
80	17
100	22
120	26
140	29
160	34
180	43
200	43
220	44
240	50
260	54
280	59
300	63
315	fracture

Testing 3

Point:	
Stress(kg)	Distance(mm)
0	0
20	9
40	13
60	21
80	28
100	36
110	45
120	53
Face:	
Stress(kg)	Distance(mm)
0	0
20	4
40	7
60	14
80	16
100	21
120	28
140	36
160	43
180	48
200	51
220	57
240	62
260	69
280	76
300	79
320	84
340	91
360	96

Line:	
Stress(kg)	Distance(mm)
0	0
20	3
40	7
60	12
80	17
100	21
120	23
140	27
160	33
180	39
200	44
220	48
240	51
260	53
280	55
300	59
320	66
340	73
356	fracture

Testing (circular tube)

Point:	
Stress(kg)	Distance(m)
0	0
20	8
40	22
60	34
80	49
100	61
120	72
150	85

Point:	
Stress(kg)	Distance(m)
0	0
20	9
40	23
60	35
80	48
100	62
120	71
150	86

Point:	
Stress(kg)	Distance(m)
0	0
20	10
40	21
60	33
80	46
100	63
120	74
150	88

Testing (square tube)

Point:	
Stress(kg)	Distance(m)
0	0
20	11
40	38
60	57
80	77
100	95
110	108
120	116

Point:	
Stress(kg)	Distance(m)
0	0
20	12
40	35
60	55
80	75
100	96
120	107
150	119

Point:	
Stress(kg)	Distance(m)
0	0
20	11
40	36
60	58
80	76
100	98
120	110
150	125

6. Capping rail

Three glass panel, each panel 2.0m maximum
 Say if middle glass panel failure, then
 horizontal load transfer to side glass (critical)
 or side glass panel failure, the horizontal load
 transfer to wall framing structure & middle glass. (not critical)

Check top rail member

$$UDL^{\#} = 0.75 \text{ kN/m} \times 2\text{m} = 1.5 \text{ kN} = 150 \text{ kg} \quad (\text{SLS}) \quad \text{critical}$$

$$P^{\#} = 0.6 \text{ kN} = 60 \text{ kg} \quad (\text{SLS})$$

From testing, the top rail can take 160 kg
 the rail is not failure (OK)
 the connection to glass panel is fixed (OK)

$$\text{Hence top rail capacity} = 160 \text{ kg} / 2\text{m} = 80 \text{ kg/m} > 75 \text{ kg/m} \quad (\text{OK})$$

Therefore use 50.8x15 RHS or 25x21 RHS as capping rail.

Check end connection to railing

Top 2x20 channels. capacity = $0.7 (2.14 \times 0.7 \times 2) = 2.1 \text{ kN}$

Design load = $0.75 \text{ kN/m} \times 1.0 \times 2.0 \text{ m} / 2 = 0.75 \text{ kN}$, Hence (OK)

or use 1mm dia screw with 30 mm penetration

MS coach screw capacity = $0.7 \times 0.7 \times (0.5 \times 30 \text{ mm} \times 2.14 \text{ mm} \times 3 \text{ mm})$
= $2.02 \text{ kN} > \text{Design load (OK)}$

Check bottom fixing

$R^* = 0.75 \text{ kN/m} \times 2 \text{ m} / 2 = 0.75 \text{ kN}$

$M^* = 0.75 \text{ kN} \times 1.0 = 0.75 \text{ kN.m}$

$T^* = 0.75 \text{ kN/m} \times 2.0 \text{ m} / 2 = 3.1 \text{ kN}$

shared by two fixings

Anchor types

① M10 Bolts channel into railpolder with 20mm embedment

$\phi T_{\text{channel}} = 13.33 \text{ kN} > T^* \text{ (OK)}$

② M10 A4-6 Bolts to steel member

$\phi T_m = 15.0 \text{ kN} > T^* \text{ (OK)}$

③ M10 Bolts with 50x50x3 washer to timber

$\phi T_m = 7.4 \text{ kN} > T^* \text{ (OK)}$

④ M10 coach screw with 30 mm penetration into wet timber or 80mm into dry timber.

$\phi T_{\text{wet}} = 3.4 \text{ kN} > T^* \text{ (OK)}$

$\phi T_{\text{dry}} = 4.8 \text{ kN} > T^* \text{ (OK)}$

7. SUMMARY

In summary, the panel & fixings tested conform to the following:

LOADS:

Live Load: For Domestic Occupancy types A and C3 (residential only) of AS/NZS 1170:2002, Table 3.3
Wind Load: HIGH as per NZS 3604, or 44 m/s Ultimate Limit State wind speed

GLASS:

Thickness: 12 mm Toughened Grade A with continuous capping rail.
Max Height: 1.13 m from the centre of upper row of fixings
Provide 50.8 OD x 1.5 mm CHS interlinking rails or 25x21mm RHS interlinking rails.

FIXINGS:

Pairs of 48 mm diameter discs at 120 mm vertical spacing

Fixing of channel to substrate to comprise any of the following:

- Pairs of M10 Dynabolts at 400 mm centres
- Pairs of M10 Chemset anchors (Maxima spin capsules or Epcon C6 series) at 400 mm centres
- Pairs of M10 G4.6 bolts to steel member at 400 mm centres
- Pairs of M10 bolts to timber (with 50 mm x 50 mm square washers) at 400 mm centres
- Pairs of M10 coachscrews at 340 mm centres with 80 mm minimum embedment into wet timber.
- Pairs of M10 coachscrews at 400 mm centres with 80 mm minimum embedment into dry timber.

Notes:

1. It is assumed that the strength and stiffness of the substrate is sufficient to adequately resist the balustrade loads - this must be confirmed for each installation situation.
2. The fixings must comply with the durability requirements on the Building Code - refer Section 4 "Durability" of NZS 3604:2011. The "Design Life" is at least 15 years.

END BUILDING
(OR 50.8 OD x 1.5 CHS POST)



CONTINUOUS CAPPING RAIL
REFER DRAWING ENG 02 & ENG 03 FOR FIXING

THIS GLASS BALUSTRADE SYSTEM IS SUITABLE FOR THE FOLLOWING:

LIVE LOAD:

WIND LOAD:

"VERY HIGH" WIND ZONE FOR BALUSTRADE, "VERY HIGH" WIND ZONES FOR POOL FENCE.

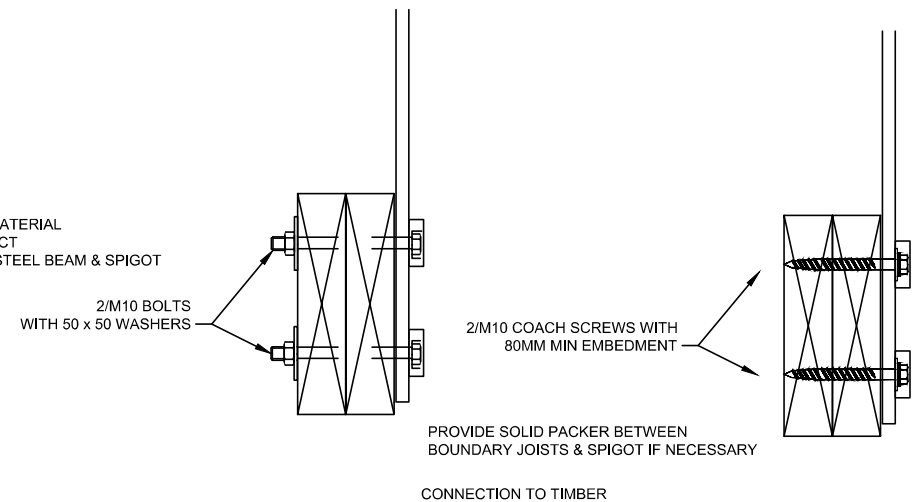
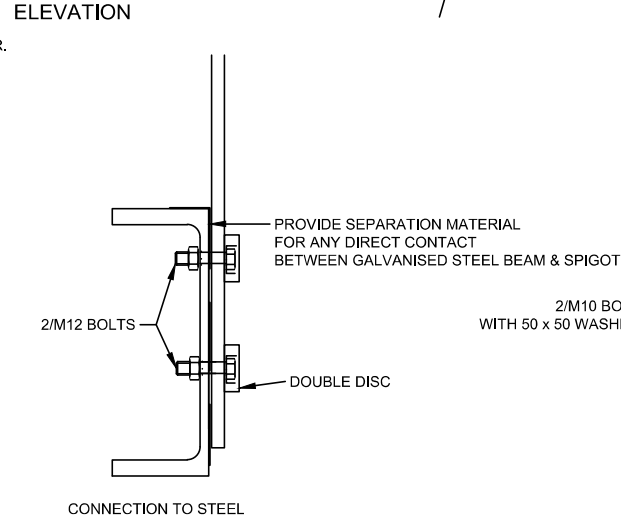
FIXINGS:

- A. ALL ANCHORS MUST BE INSTALLED IN ACCORDANCE WITH MANUFACTURERS REQUIREMENTS.
B. BOLTING ARRANGEMENT MUST COMPLY WITH RELEVANT MATERIAL CODES.
C. STRENGTH & STIFFNESS OF THE SUBSTRATE STRUCTURE MUST BE SUFFICIENT FOR THE BALUSTRADE LOADS.

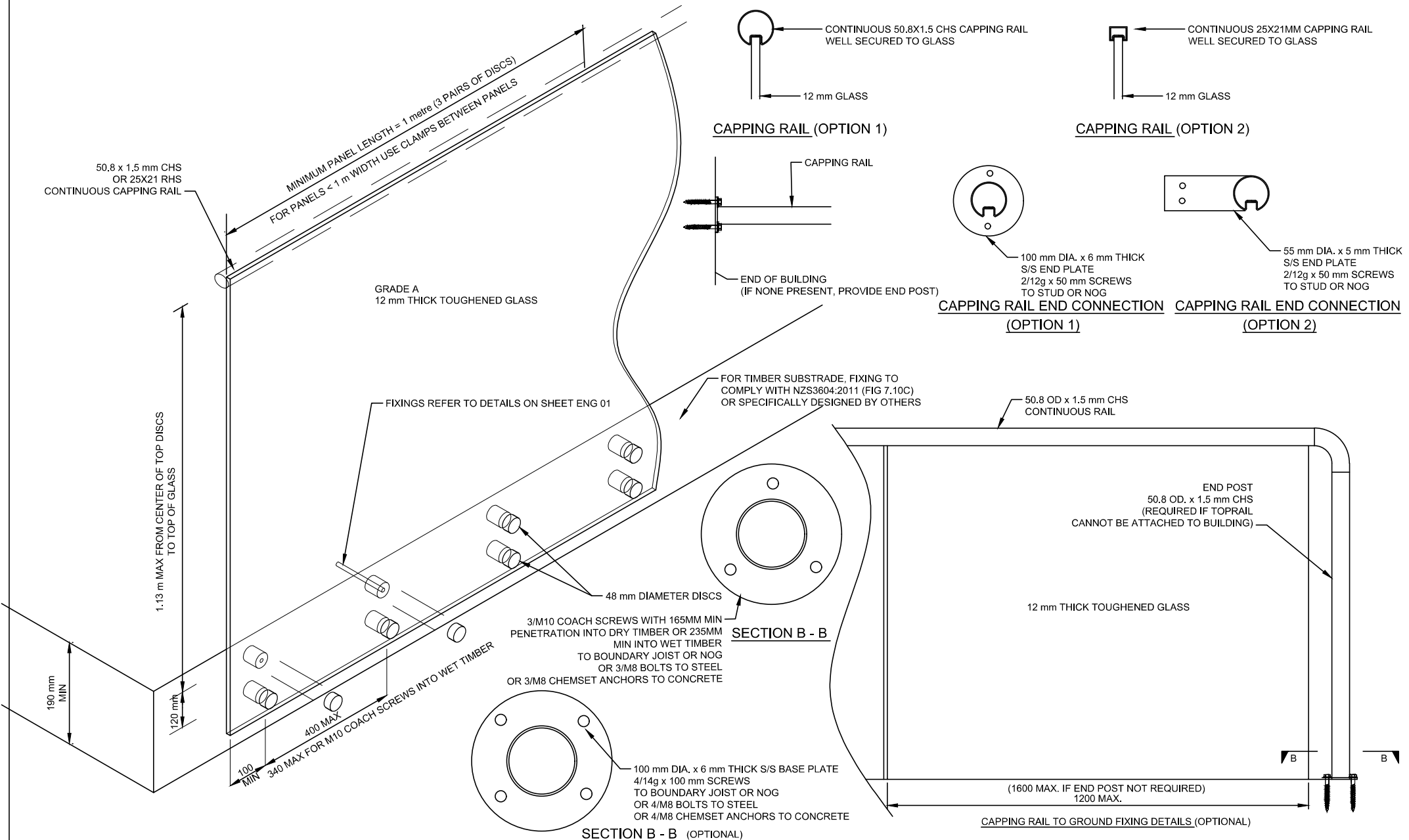
- 1.M10 CHEMSET ANCHORS (MAXIMA SPIN CAPSULES OR EPCON C6 SERIES) OR M10 DYNABOLT ANCHORS TO CONCRETE. THE ANCHORS MUST COMPLY FULLY WITH THE MANUFACTURER'S REQUIREMENTS.
- 2.M10 G4.6 BOLTS TO STEEL MEMBERS. THE BOLTING ARRANGEMENT MUST COMPLY WITH NZS 3404:1997.
- 3.M10 G4.6 BOLTS TO TIMBER WITH 50 mm x 50 mm SQUARE WASHERS. THE BOLT ARRANGEMENT MUST COMPLY WITH NZS 3603:1993.
- 4.M10 COACH SCREWS TO DRY TIMBER, 80 mm MIN. EMBEDMENT
- 5.M10 COACH SCREWS TO WET TIMBER, 80 mm MIN. EMBEDMENT, MAX. DISC SPACING 340 mm.

DURABILITY

THE FIXINGS MUST COMPLY WITH SECTION 4 "DURABILITY" OF NZS 3604:2011

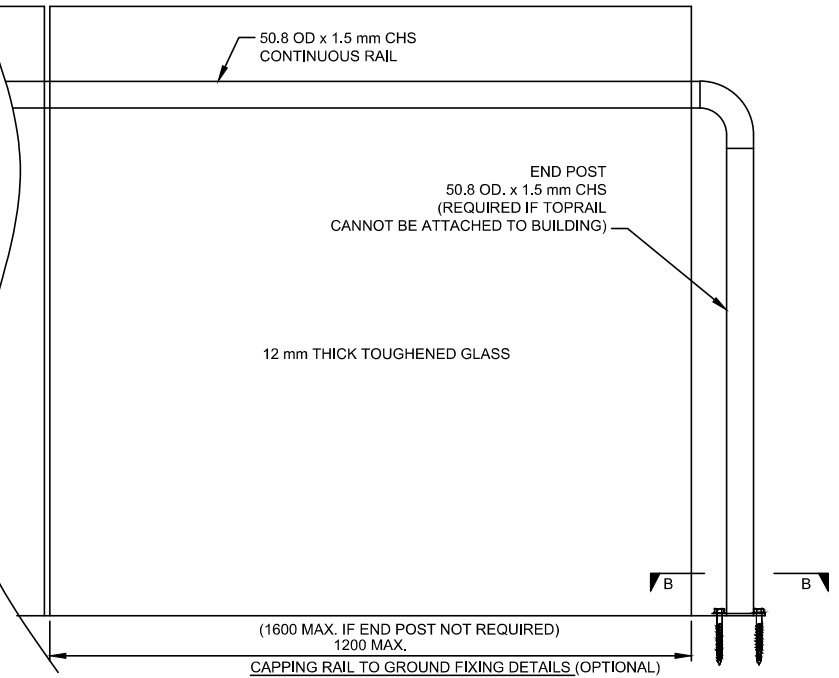
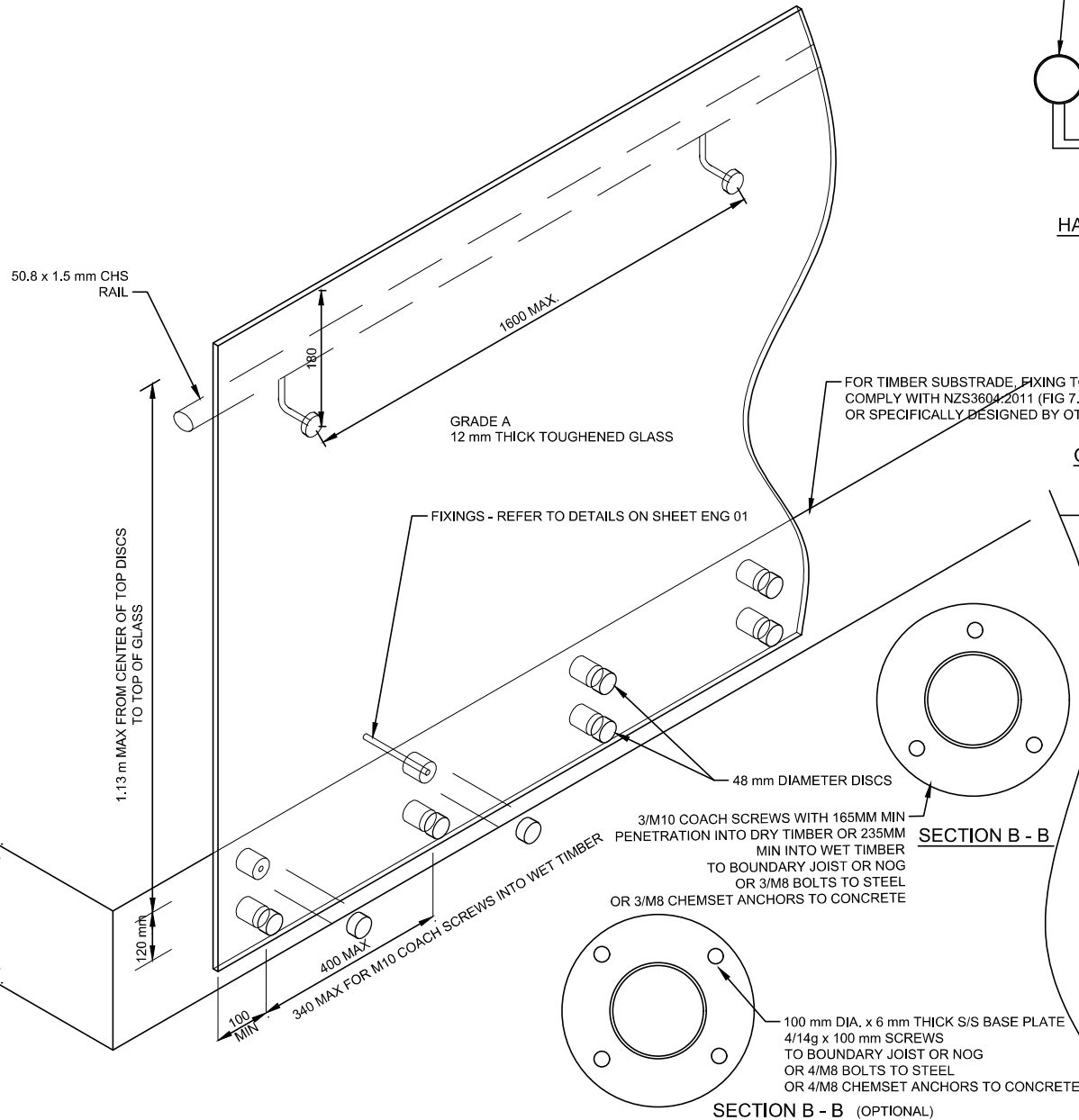
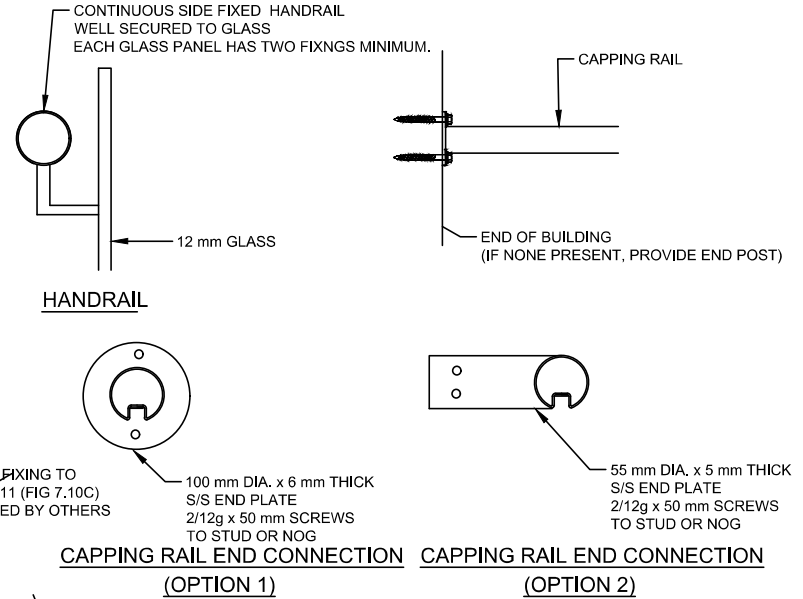


FRAMELESS GLASS BALUSTRADES SUPPORTED BY DOUBLE DISCS

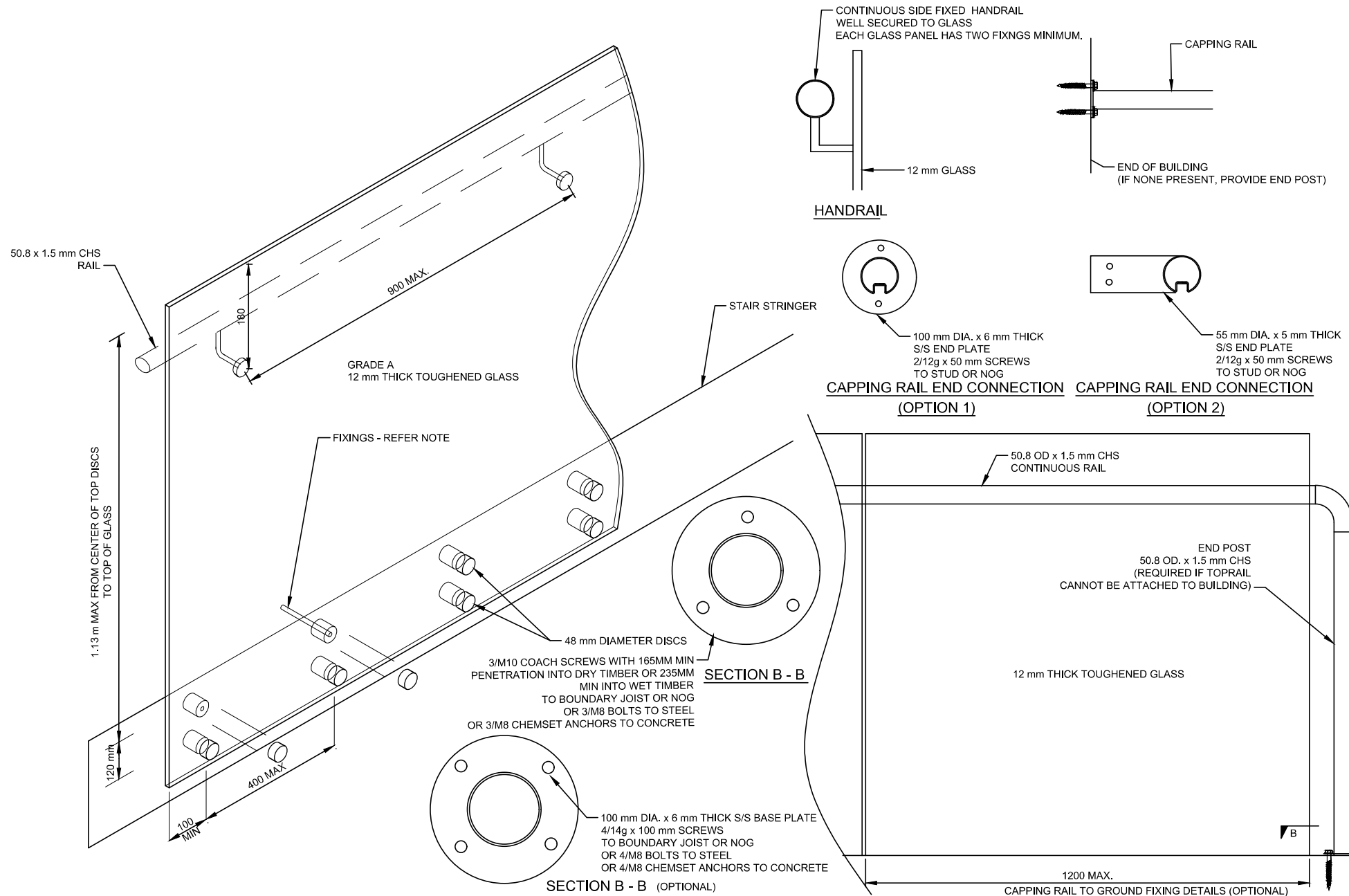


P & P CONSULTING ENGINEERS LTD Civil and Structural Engineering 6A Montel Avenue, Henderson ph: 836-1853	ROYAL GLASS LTD FRAMELESS GLASS BALUSTRADES		Scales 1:10 (A3)		Project 20/059	
	DOUBLE DISC SYSTEM		Drawn HY	Date 25/06/20	Sheet ENG 02	

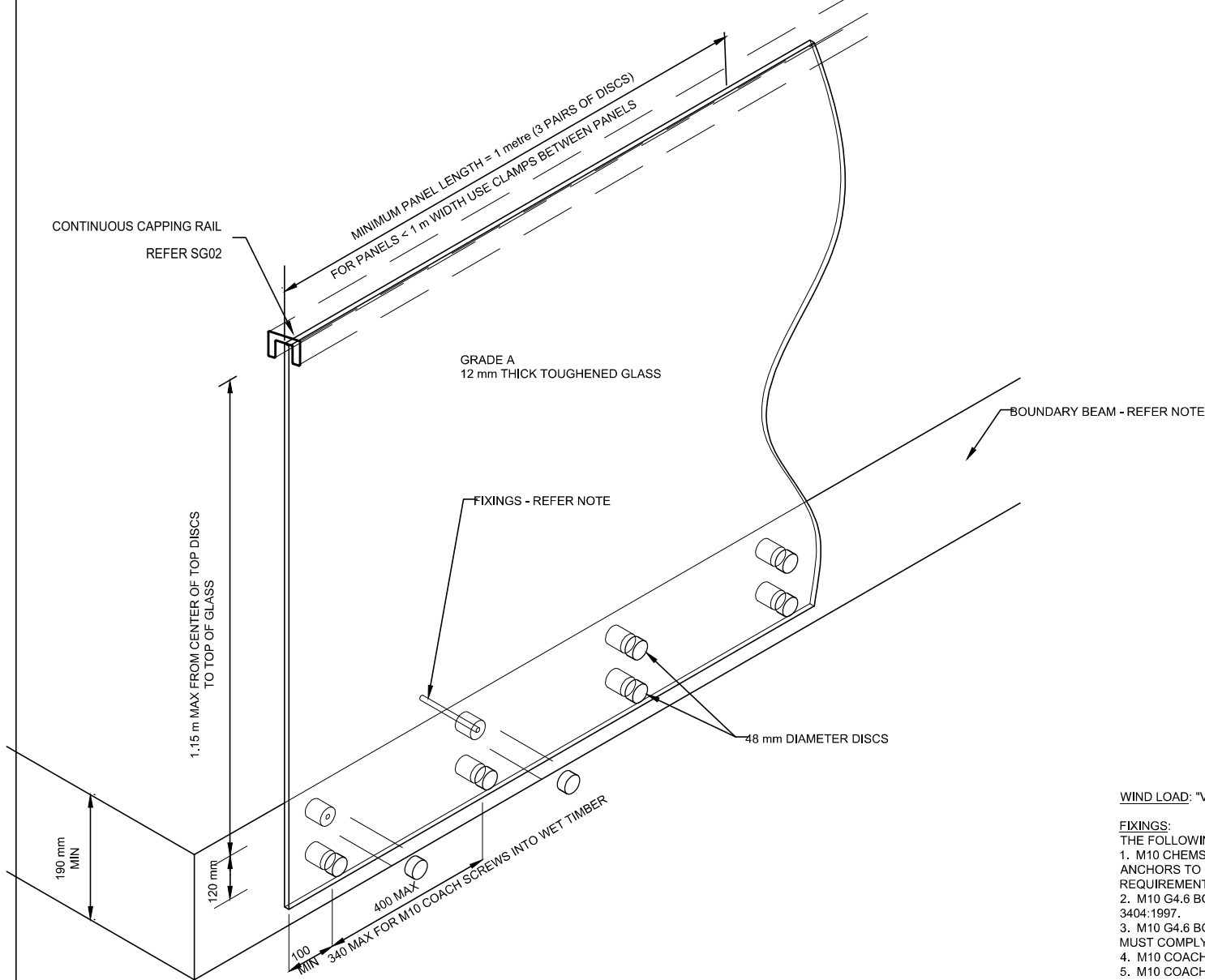
FRAMELESS GLASS BALUSTRADES SUPPORTED BY DOUBLE DISCS



FRAMELESS GLASS BALUSTRADES SUPPORTED BY DOUBLE DISCS



FRAMELESS GLASS BALUSTRADES SUPPORTED BY DOUBLE DISCS



WIND LOAD: "VERY HIGH" WIND ZONE WITH BASIC WIND SPEED UP TO 50 m/s.

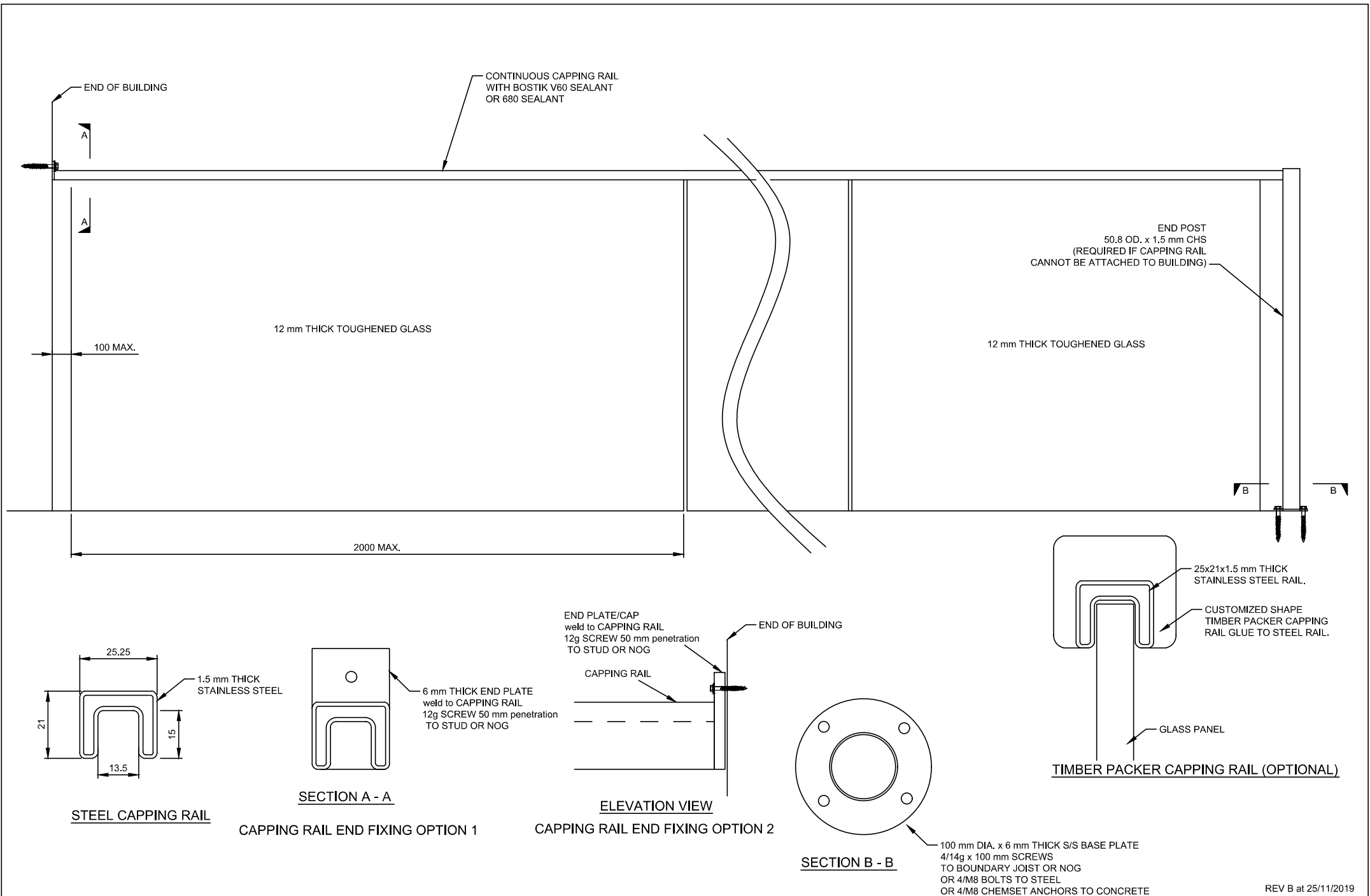
FIXINGS:

THE FOLLOWING FIXINGS CAN BE USED:

1. M10 CHEMSET ANCHORS (MAXIMA SPIN CAPSULES OR EPCON C6 SERIES) OR M10 DYNABOLT ANCHORS TO CONCRETE. THE ANCHORS MUST COMPLY FULLY WITH THE MANUFACTURER'S REQUIREMENTS.
2. M10 G4.6 BOLTS TO STEEL MEMBERS. THE BOLTING ARRANGEMENT MUST COMPLY WITH NZS 3404:1997.
3. M10 G4.6 BOLTS TO TIMBER WITH 50 mm x 50 mm SQUARE WASHERS. THE BOLT ARRANGEMENT MUST COMPLY WITH NZS 3603:1993.
4. M10 COACH SCREWS TO DRY TIMBER, 80 mm MIN. EMBEDMENT
5. M10 COACH SCREWS TO WET TIMBER, 130 mm MIN. EMBEDMENT, MAX, DISC SPACING 345 mm.

BOUNDARY BEAM

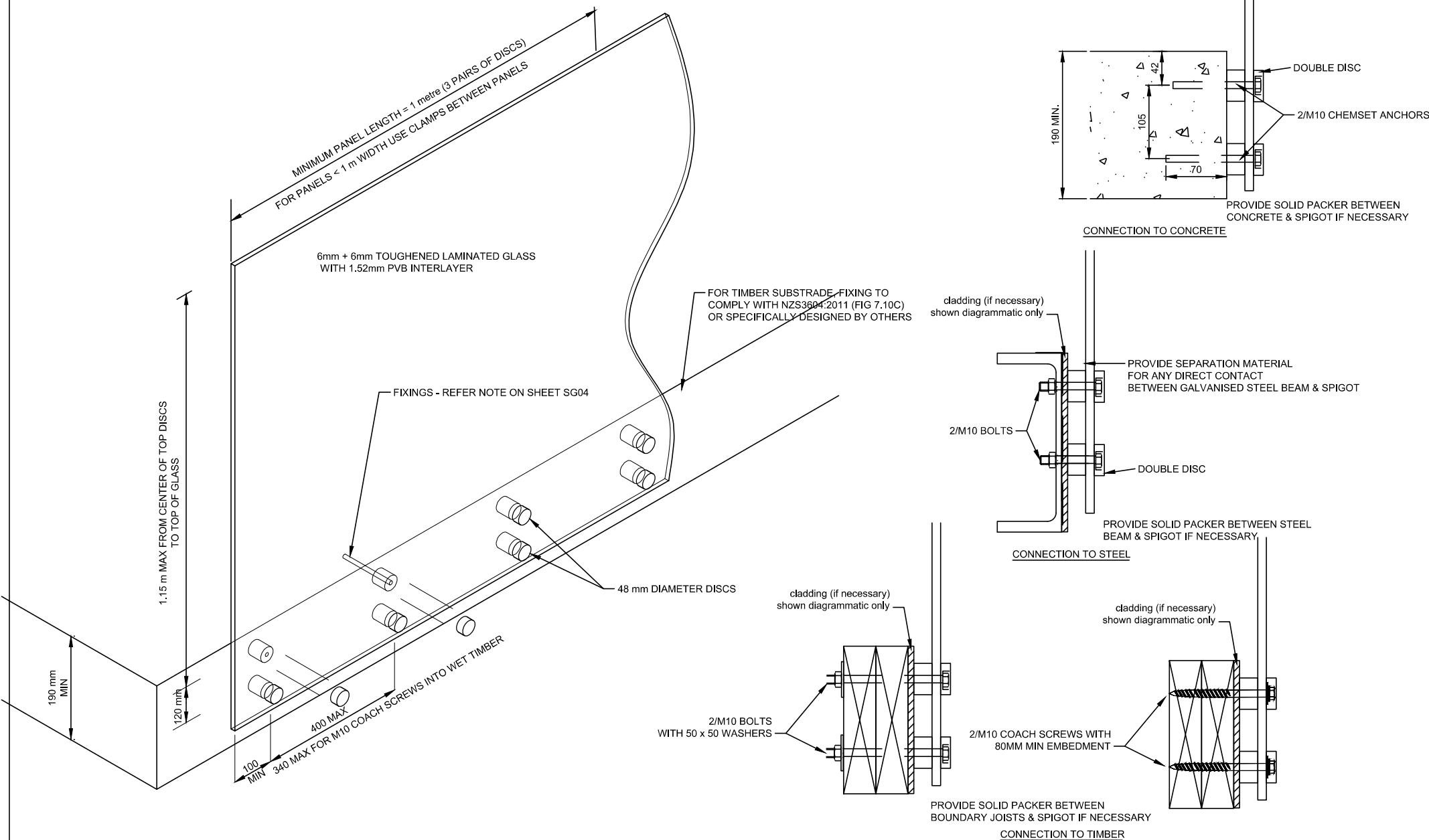
BOUNDARY BEAM & FIXING TO COMPLY WITH EITHER NZS3604:2011 FIG 7.10C OR SPECIFICALLY DESIGNED BY OTHERS.



P & P CONSULTING ENGINEERS LTD Civil and Structural Engineering 6A Montel Avenue, Henderson ph: 836-1853	ROYAL GLASS LTD FRAMELESS GLASS BALUSTRADES		Scales NTS		Project 20/059
	CAPPING RAIL & FIXING DETAIL		Drawn PP	Date 25/06/20	Sheet ENG 06

REV B at 25/11/2019

FRAMELESS LAMINATED GLASS BALUSTRADES SUPPORTED BY DOUBLE DISCS



P & P CONSULTING ENGINEERS LTD Civil and Structural Engineering 6A Montel Avenue, Henderson ph: 836-1853	ROYAL GLASS LTD FRAMELESS GLASS BALUSTRADES		Scales 1:10 (A3)		Project 20/059	
	DOUBLE DISC SYSTEM		Drawn HY	Date 25/06/20	Sheet ENG 07	



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